Tattva: Product Requirements Document

THE IMMUTABLE LEDGER OF HUMAN COGNITIVE LABOR

> **The Philosophical Core: Friction as a Feature**
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> The modern internet is fundamentally optimized for the elimination of friction. Platforms measure success by how effortlessly a user can consume information, leading to the "infinite scroll" and the deterioration of deep attention. Tattva operates on the inverse principle: *Friction is the product*. By systematically introducing cognitive roadblocks, unyielding technical constraints, and adversarial AI evaluation, Tattva transforms passive learning into a verifiable intellectual crucible. We are not building a social network; we are building the Bloomberg Terminal for human capital, replacing the traditional university transcript with a cryptographic, un-fakeable proof of deep execution.

1. Executive Summary & Market Positioning

In the epoch of generative AI, the value of the traditional university degree is collapsing. When an AI agent can write an essay, generate a codebase, or solve a physics examination in seconds, the output of a student is no longer a reliable metric of their competence. Enterprise recruiters at elite engineering firms (such as Google, Palantir, and Tesla) are facing an unprecedented signal-to-noise crisis: resumes are flawless, but underlying candidate capability is hollow.

Tattva solves this structural market failure by shifting the metric of evaluation from *output* to the *process of inquiry*. Tattva is a B2B SaaS credentialing engine disguised as an elite, high-friction learning platform. We target the most fiercely competitive segment of the academic market—GATE , ISRO aspirants and computer science undergraduates—providing them with a platform that forces deep engagement with dense technical material (like state space models, compiler design, and memory safety). Every action a student takes on the platform is logged into an append-only ledger, generating a "Cognitive Stamina Score" and a public Knowledge Graph that visually and mathematically proves their capability to enterprise recruiters.

2. Feature Deep Dive: The 25-Minute Baseline Constraint

The Problem Being Solved

The "Reel" and "Short" formats have conditioned users to expect immediate, dopamine-driven resolutions to complex topics. This format completely destroys the nuance required for high-level systems engineering. A creator cannot accurately explain out-of-order CPU execution or distributed consensus algorithms in 60 seconds without abstracting away the exact mechanical truths that actually matter. The platform must structurally prevent the dilution of knowledge by imposing a strict chronological floor.

Functional Implementation & Mechanics

Tattva enforces a hard, unyielding constraint at the ingestion layer: no media, lecture, or document embed can be processed if its total runtime (or reading time equivalent) is under 25 minutes. If a creator attempts to upload or link a 14-minute tutorial, the backend immediately rejects the request. This 25-minute floor deliberately aligns with the standard "Pomodoro" deep-work cycle. It serves as a formidable barrier to entry, naturally filtering out content farmers, engagement baiters, and superficial summarizers. For the user, this constraint establishes immediate psychological context. When a student opens a Tattva node, they know unequivocally that they are committing to at least a half-hour of rigorous cognitive labor. This eliminates the "just one more video" slot-machine effect and replaces it with deliberate, scheduled intellectual execution.

3. Feature Deep Dive: The Knowledge Scaffold (Semantic Tree)

The Problem Being Solved

Traditional AI wrappers (like ChatGPT or NotebookLM) are inherently ephemeral and linear. A student asks a question, receives an answer, and the conversation vanishes into a vertical scroll. If that student spends three hours deeply dissecting a 40-page academic paper on memory arena allocators, their labor is invisible. Furthermore, standard chat interfaces encourage "single-player" learning, where breakthroughs are siloed and subsequent students at the same university must redundantly start from scratch.

Functional Implementation & Mechanics

Tattva completely abandons the standard chat interface. Instead, the UI is split 50/50. The left pane holds the dense source material (the embedded 25-minute video or parsed technical PDF). The right pane houses the **Knowledge Graph**, rendered dynamically using React Flow. When a user asks an initial question, it generates the "Root Node." The Sarvam AI backend processes the transcript chunk, provides an explanation, and—crucially—generates exactly three adaptive, thought-provoking follow-up meta-questions. When the user selects a follow-up, a new "Child Node" branches out visually on the screen. Over a two-hour session, the student organically constructs a massive, interconnected tree of logic. This graph is entirely persistent and shareable. A peer can click on the student's public link, bypass the initial introductory nodes, and immediately branch their own inquiries off the advanced edges of the original student's tree. This creates a compounding, multiplayer network of high-density knowledge.

4. Feature Deep Dive: The "Tattva 200" (The Execution Gauntlet)

The Problem Being Solved

Open-ended learning platforms suffer from the "blank canvas" paralysis. A new user stares at an empty search bar and abandons the platform because they do not know what they do not know. Furthermore, unstructured learning is difficult to measure and credentialize. To capture the massive, highly motivated market of competitive exam aspirants (specifically GATE CS students), the platform must provide a ruthless, highly structured path of execution.

Functional Implementation & Mechanics

Inspired by the famous "Blind 75" list used for LeetCode software engineering interviews, the **Tattva 200** is the definitive, platform-curated execution gauntlet for Computer Science and Electronics. It is presented on the user's home dashboard as a massive grid of locked nodes, representing the 200 most brutally difficult, non-negotiable concepts required to master the discipline (e.g., Turing Reducibility, Page Fault Handling, B-Tree rebalancing, and Cache Coherence Protocols). Users cannot skip ahead; nodes unlock sequentially based on prerequisite completion. This gamifies the academic curriculum, turning abstract syllabus requirements into a tangible, visually satisfying conquest map. It provides instant utility to the student by structuring their daily study routine, while providing Tattva with a standardized baseline to compare different candidates. If a recruiter sees that a candidate has fully conquered the Tattva 200, they have absolute, verified mathematical proof of that candidate's foundational competence.

5. Feature Deep Dive: "Be Careful Here!!" (Adversarial AI Traps)

The Problem Being Solved

Current Large Language Models are designed to be excessively servile and "helpful." If a student asks a leading question containing a fundamental misunderstanding, standard AI will often politely answer the question without explicitly correcting the underlying false premise. This breeds the "illusion of competence," where a student feels productive but would immediately fail a technical interview when a senior engineer pushes back on their assumptions.

Functional Implementation & Mechanics

Tattva introduces pedagogical friction through an Adversarial AI framework. Our system prompts instruct the Sarvam backend to actively scan the user's input against known, common misconceptions within the specific discipline. If a student is studying multithreading and asks a question that implies a misunderstanding of race conditions, the AI temporarily halts the standard Q&A flow. The UI flashes a high-visibility, high-contrast alert component reading: **"⚠️ Be Careful Here!!"** The AI then presents a specific, dangerous edge-case scenario (a "Trap") derived from their misunderstanding (e.g., a potential deadlock scenario in their logic). The standard chat input is disabled, and the student is forced to explicitly solve or explain the trap before the graph allows them to proceed. This mimics the exact pressure of a real-world technical interview, ensuring that the knowledge built on the platform is battle-tested and structurally sound.

6. Feature Deep Dive: "Did You Conquer It?" (Verified High-Stakes Badges)

The Problem Being Solved

In order for Tattva to scale with zero marketing capital, the platform requires an intrinsic viral loop. Users need a compelling, ego-driven reason to share their Tattva profiles on external networks like LinkedIn, GitHub, and Twitter. Traditional certificates of completion are meaningless because they simply prove a user clicked "Next" on a video player.

Functional Implementation & Mechanics

Tattva introduces "Boss Fight" nodes. Within the Tattva 200 curriculum, certain critical nodes are algorithmically flagged as having historically low completion rates (e.g., a sub-5% pass rate for a deep dive into the Mathematics of Cryptography). When a user navigates one of these nodes, the AI does not just answer questions; it demands a comprehensive, first-principles synthesis of the entire concept without relying on analogies. If the Sarvam API evaluates the student's final explanation as passing the density and accuracy threshold, the UI triggers a massive, satisfying visual celebration and permanently attaches a cryptographic **"Conquered"** badge to the node on the user's public transcript. This badge is incredibly difficult to earn, making it highly prestigious. The user is immediately prompted to share this specific victory on LinkedIn. When their network clicks the link, they do not see a generic certificate; they see the actual, massive Knowledge Graph the user built to earn the badge, immediately proving Tattva's value to hundreds of potential new users.

7. Feature Deep Dive: "Research Mode" (The Training Weights)

The Problem Being Solved

As students progress from beginners to advanced systems engineers, their need for direct hand-holding decreases, but their need for rigorous sparring increases. If the AI simply gives the user the answer every time they are stuck on an advanced dynamic programming algorithm, it destroys the necessary cognitive struggle required for actual mastery. The platform must scale its difficulty dynamically.

Functional Implementation & Mechanics

The UI features a prominent, physical-looking toggle switch labeled **[Research Mode]**. When activated by the user, the entire interface shifts to a high-contrast dark theme, visually signaling a change in the environment's rules of engagement. Under the hood, toggling Research Mode drastically alters the system prompt sent to the Sarvam API. The AI is now strictly forbidden from outputting direct answers, code snippets, or simplified summaries. Instead, it acts as a ruthless academic mentor. If the user asks a question, the AI will only respond with raw academic citations, links to official documentation (e.g., the C++ standard reference), or Socratic counter-questions that force the user to derive the answer themselves. Because this mode requires immensely more cognitive labor, the database applies a **3x Multiplier** to the user's Cognitive Stamina Score for every node completed while the mode is active. This incentivizes top-tier students to take the harder path, proving their resilience.

8. Feature Deep Dive: "Throw the Best Explanation" (The Arena)

The Problem Being Solved

Creating high-density seed content (transcripts, base explanations, foundational graphs) is capital intensive. If Tattva relies solely on the founding team to generate the base material for all 200 CS concepts, development will bottleneck immediately. We need a mechanism to crowdsource the highest quality content directly from the smartest users on the platform without paying them cash.

Functional Implementation & Mechanics

We solve the content pipeline through decentralized, ego-driven bounties called "The Arena." Every week, Tattva posts a challenge on a notoriously abstract topic (e.g., "Explain the mathematical intuition behind Fourier Transforms in Signal Processing"). Instead of the AI generating the baseline graph, the community is invited to build their best, most rigorous explanation graph. Users submit their completed graphs to the Arena, where they are peer-reviewed by the community. The voting mechanism is heavily weighted; votes from users with high Stamina Scores count significantly more than votes from new users, preventing gamification. The winning submission is permanently crowned as the official "Seed Graph" for that topic across the entire Tattva platform. The creator receives a massive Stamina bounty and permanent attribution. We effectively crowdsource our core database from the top 1% of engineering students by paying them in prestige and visibility.

9. Feature Deep Dive: Standardized Visual Architecture

The Problem Being Solved

The architecture of a platform dictates the behavior of its users. If you allow custom image thumbnails, you inevitably create an arms race for attention. Creators will inevitably devolve into using bright colors, emotive faces, and deceptive clickbait typography to win the click, completely destroying the academic, rigorous atmosphere necessary for deep learning.

Functional Implementation & Mechanics

Tattva outright bans image uploads. When a creator (or the system) generates a new video node or topic, they are forced to use the Tattva CSS-Grid Thumbnail Generator. The platform provides a strictly limited palette of 5 minimalist, typographic templates inspired by vintage academic journals, university press book covers, and terminal interfaces. The creator inputs structured metadata: the Title, the Primary Interlocutor, the Core Subject, and the Learning Intent (e.g., Research, Postmortem, Synthesis). The system renders this metadata perfectly into the CSS grid. This guarantees that every single piece of content on the platform looks uniformly professional, intellectual, and clean. It signals to the user that the value lies entirely in the density of the thought, not the flashiness of the packaging.

10. Feature Deep Dive: The Stamina Ledger & B2B Telemetry

The Problem Being Solved

Consumer social networks fail when they try to pivot to education because their core metric—the "View" or the "Like"—is economically worthless to a B2B client. A recruiter does not care how many "Likes" a student got on a study post. They care about verified, un-fakeable hours of deep work. Tattva must monetize by providing data that enterprises cannot get anywhere else.

Functional Implementation & Mechanics

The backend of Tattva operates an append-only ledger in PostgreSQL (via Supabase) called `WalletTransactions`. This is the heart of the B2B engine. We track "Cognitive Stamina." Every time a user initiates a study session, ignores a direct answer, selects a deep-dive meta-question, survives a "Be Careful Here!!" trap, or operates in Research Mode, the ledger securely writes an immutable transaction awarding them Stamina points. Because this logic is executed entirely server-side, it cannot be spoofed by the client. Over time, a student builds a massive Stamina Score that represents hundreds of hours of verified intellectual labor. Tattva monetizes by selling API access and dashboard seats to enterprise recruiters. A recruiter at a top firm can query the Tattva database: *"Show me all computer science students in Bengaluru who have conquered the Out-of-Order Execution node, operated primarily in Research Mode, and hold a Stamina Score in the top 5th percentile."* We bypass the traditional resume entirely, providing a direct pipeline to the most resilient, capable engineers in the market.

11. Hackathon Execution Plan & Team Architecture (Nov 13-14)

To win the ₹13,000 first prize and deploy the 500 Sarvam API credits effectively during the 24-hour hackathon window at REVA University, the 4-person team must adhere to a strict, ruthless "Golden Path" MVP strategy. We will not build the full marketplace; we will build the "Crucible."

Team Member Role	Hackathon Responsibilities & Focus Areas
Member 1: UI & Gamification Architect	Develops the Split-Screen `Tattva Workspace`. Integrates `react-flow-renderer` for the visual Knowledge Tree. Builds the "Research Mode" toggle and the high-visibility "Be Careful Here!!" adversarial UI trap components.
Member 2: Sarvam API Integration Lead	Constructs the Node.js API bridge. Designs the complex system prompts that force the Sarvam LLM to output rigid JSON containing the explanation, 3 follow-ups, and the critical `is_trap` boolean flag to trigger adversarial UI states.

Team Member Role	Hackathon Responsibilities & Focus Areas
Member 3: Supabase Schema & Ledger Master	Designs the relational database. Implements `Inquiries`, `QuestionNodes`, and the append-only `WalletTransactions` ledger. Ensures Research Mode correctly writes the 3x Stamina multiplier securely from the server side.
Member 4: Product & Demo Director	Curates the seed content (e.g., a dense transcription of a Memory Allocation lecture). Writes the exact pitch script. Controls the live demo flow to ensure the AI hits the trap state flawlessly in front of the judges without runtime errors.

The Live Demo Strategy (The Kill-Shot)

When presenting on November 14th, the pitch must immediately reframe the judges' expectations. Do not pitch an "Ed-Tech App." Pitch a **Credentialing Engine**. Walk the judges through the "Tattva 200" dashboard. Click on a notoriously difficult CSE concept. Toggle "Research Mode" on. Speak a complex question into the Sarvam voice API. When the AI detects a misconception and halts the UI with the "Be Careful Here!!" trap, the room will instantly recognize the paradigm shift. You are not building a tool that makes learning easy; you are building a tool that makes learning rigorous, measurable, and economically valuable.

> **Final Directive**
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> Tattva succeeds only if we refuse to compromise on friction. If we make the platform easier, we destroy the signal. If we destroy the signal, we lose the enterprise recruiters. Stick to the 25-minute minimums, enforce the CSS templates, and let the AI challenge the user ruthlessly. The market is starving for proof of actual competence. We are building the infrastructure to provide it.